Solid State and Devices

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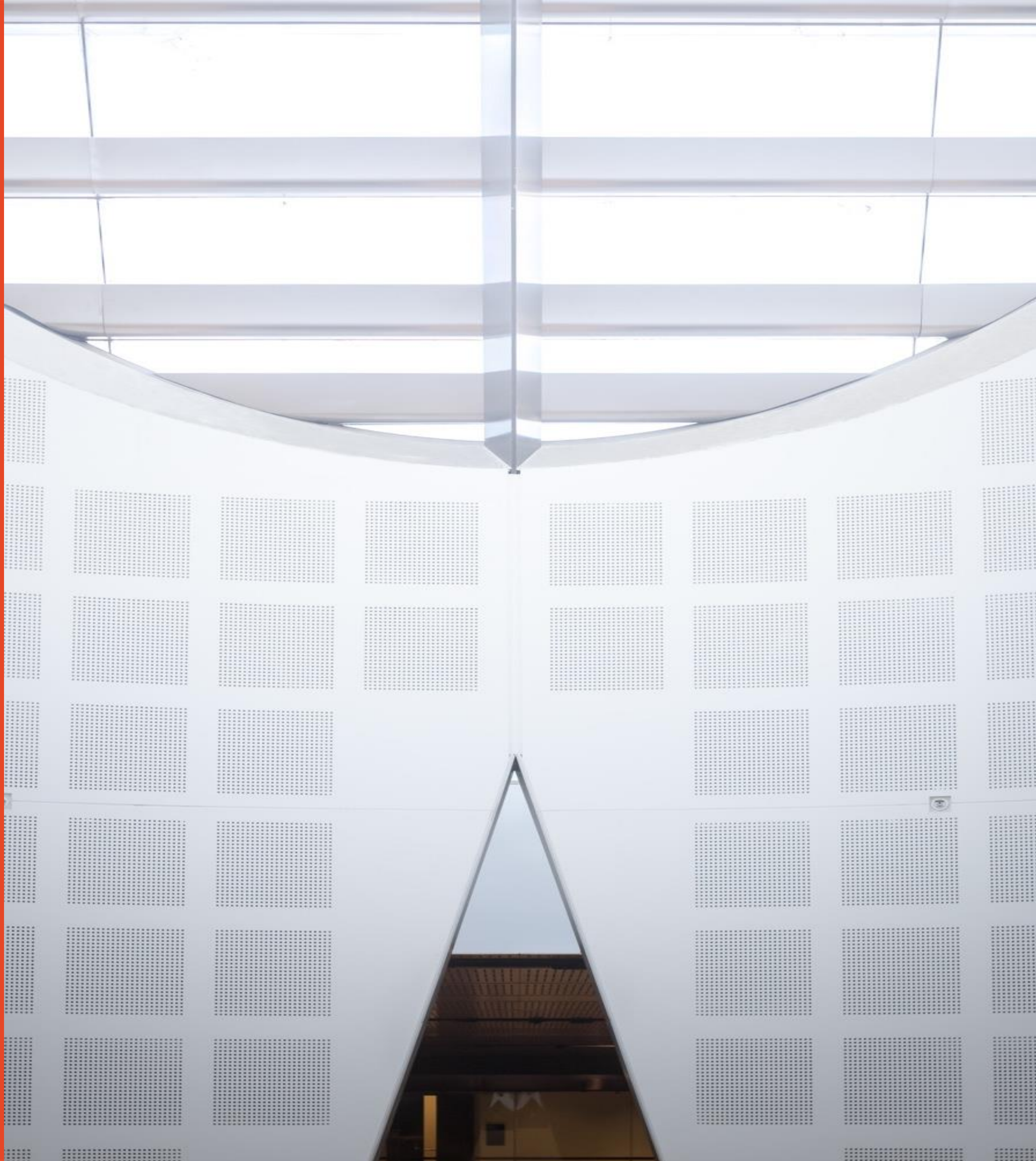


Lesson 4

- Pauli principle

- **Principle:** No two identical fermions can occupy the same quantum state.
- **Mathematically:** The total wavefunction of identical fermions is **antisymmetric** under particle exchange:
- **(a) Indistinguishability of identical particles**
- Identical particles are fundamentally indistinguishable.
- Exchanging particle labels cannot change physical predictions.
- This means the total state under exchange can only pick up a **phase factor**:

Identical particles are indistinguishable: exchanging particle labels cannot alter physical predictions.

Therefore, upon exchange of two particles, the many-particle wavefunction may only change by a phase factor:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = e^{i\phi}\, \Psi(\mathbf{r}_2, \mathbf{r}_1).$$

.

Applying the exchange operator twice gives

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = e^{i2\phi}\, \Psi(\mathbf{r}_1, \mathbf{r}_2).$$

Hence $e^{i2\phi} = 1 \implies e^{i\phi} = \pm 1.$

This yields two possible classes of quantum statistics:

- Symmetric wavefunctions $(+)$: **Bosons**.
- Antisymmetric wavefunctions $(-)$: **Fermions**.

- Particles with *integer spin* $(0, 1, 2, \dots)$ are bosons, obeying symmetric wavefunctions.
- Particles with *half-integer spin* $(1/2, 3/2, \dots)$ are fermions, obeying antisymmetric wavefunctions.

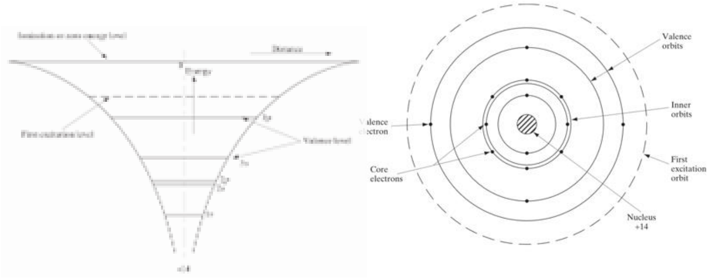
Physical Implications

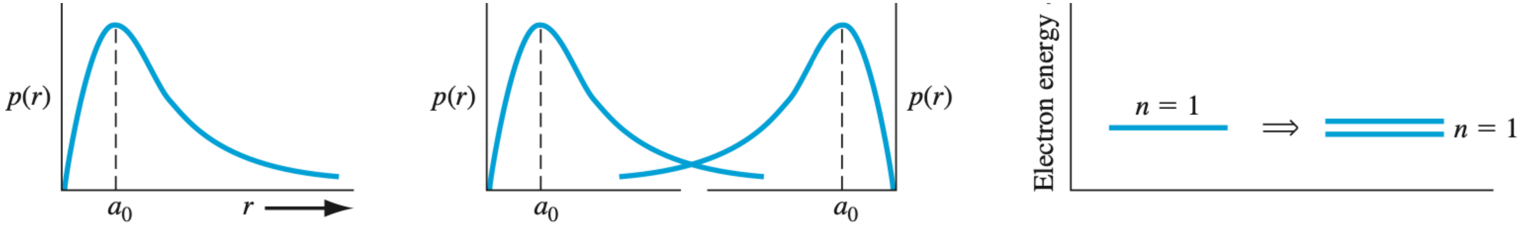
- Fermions: electrons, protons, neutrons. Their antisymmetric wavefunctions forbid double occupation of the same quantum state, giving rise to the structure of the periodic table, electron degeneracy pressure, and many features of condensed matter.
- Bosons: photons, phonons, ^{4}He atoms. Their symmetric wavefunctions allow many particles in the same state, enabling lasers and Bose–Einstein condensates.

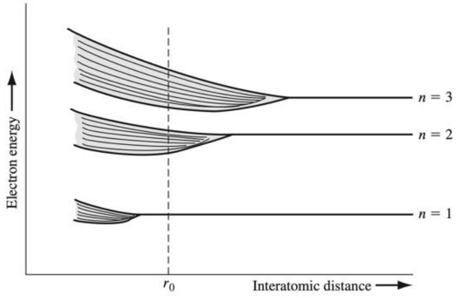
1. Indistinguishability $\Rightarrow$ wavefunctions must be symmetric or antisymmetric.
2. Spin–statistics theorem $\Rightarrow$ half-integer spin $\rightarrow$ fermions (antisymmetric), integer spin $\rightarrow$ bosons (symmetric).
3. For fermions, antisymmetry immediately enforces the Pauli exclusion principle: two fermions cannot occupy the same state.

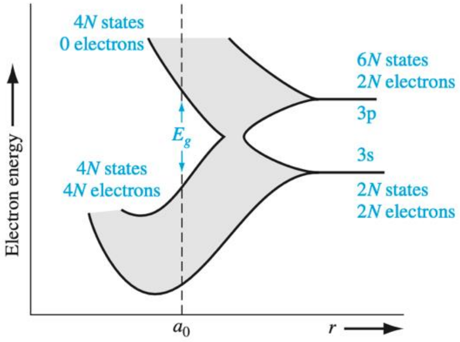
When many identical atoms are brought together to form a solid:

- The outer (valence) electrons experience overlap between atomic orbitals of neighbouring atoms.
- According to the Pauli exclusion principle, no two electrons can occupy the same quantum state. As a result, the degeneracy of atomic levels is lifted, and the single atomic level splits into N closely spaced levels, where N is the number of atoms.
- In a macroscopic solid ($N \sim 10^{23}$), these levels are so closely spaced that they form a *quasi-continuum* called an **energy band**.

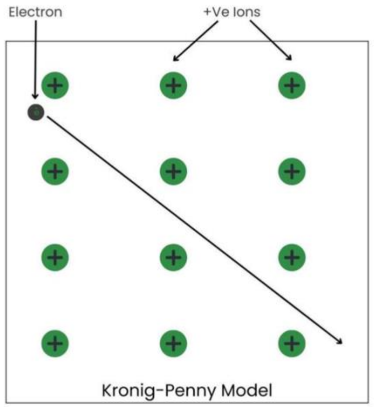






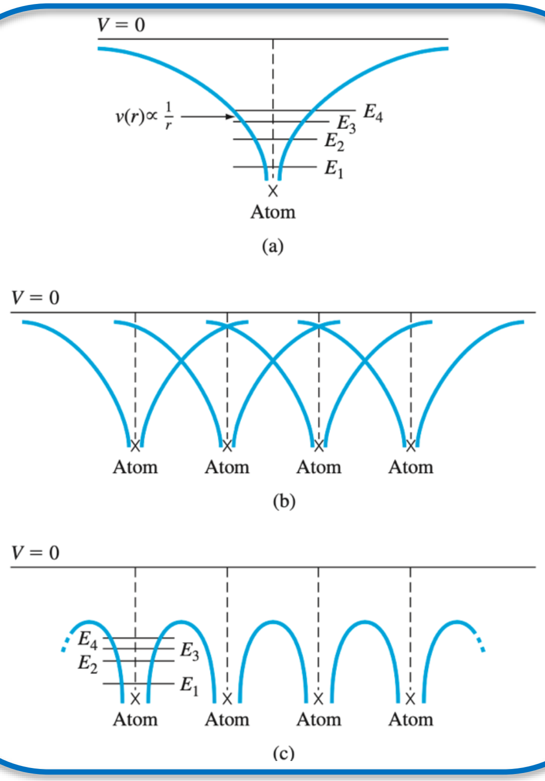


Periodic Lattice



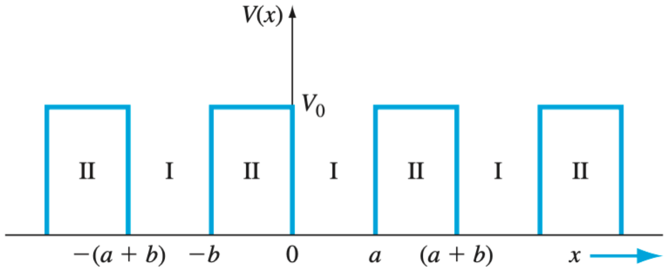
$$V(x+d) = V(x)$$

$$d = a + b:$$



Simplified Model

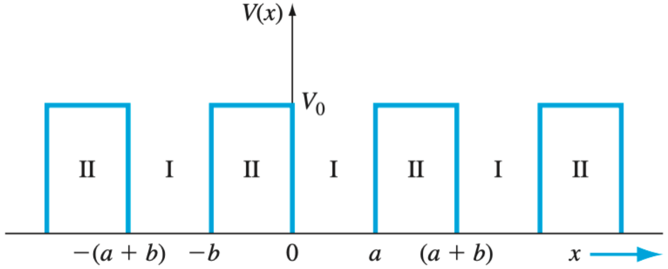
$$V(x) = \begin{cases} 0, & -b < x < 0, \\ V_0, & 0 < x < a, \end{cases}$$



Block Theorem (in periodic potentials)

$$\psi_k(x) = e^{ikx} u_k(x), \quad u_k(x+d) = u_k(x).$$

$$V(x) = \begin{cases} 0, & -b < x < 0, \\ V_0, & 0 < x < a, \end{cases}$$



In any periodic potential:

$$\psi_k(x) = e^{ikx} u_k(x), \quad u_k(x+d) = u_k(x)$$

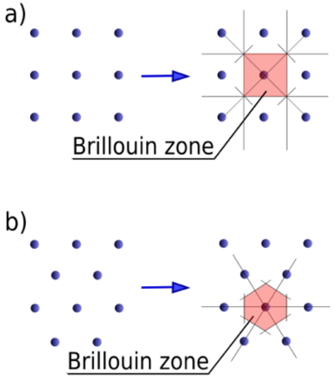
Here k is the wavevector, and $u_k(x)$ is periodic with the lattice.

$$G_n = \frac{2\pi n}{d}, \quad n \in \mathbb{Z}$$

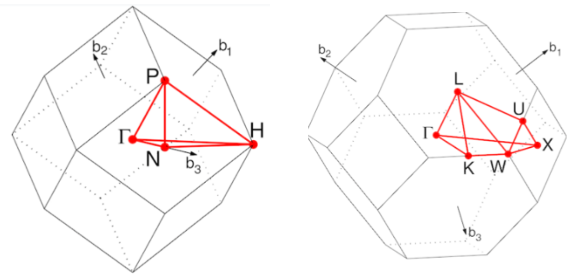
1st Brillouin zone: $\quad -\frac{\pi}{d} \leq k < \frac{\pi}{d}$

In mathematics and solid state physics, the first Brillouin zone is a uniquely defined primitive cell in reciprocal space.

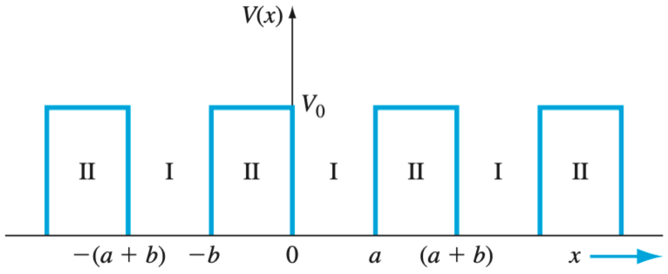
2D



3D



$$V(x) = \begin{cases} 0, & -b < x < 0, \\ V_0, & 0 < x < a, \end{cases}$$



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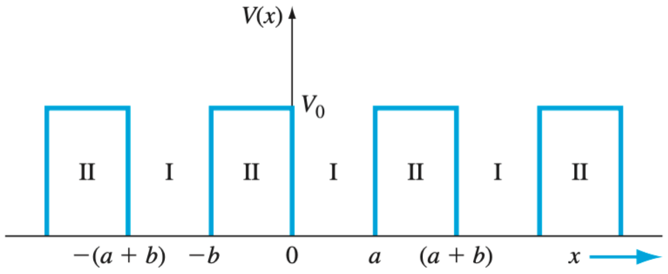
Here k is the wavevector, and $u_k(x)$ is periodic with the lattice.

$$\frac{d\psi}{dx} = \frac{du}{dx} e^{ikx} + iku(x)e^{ikx},$$

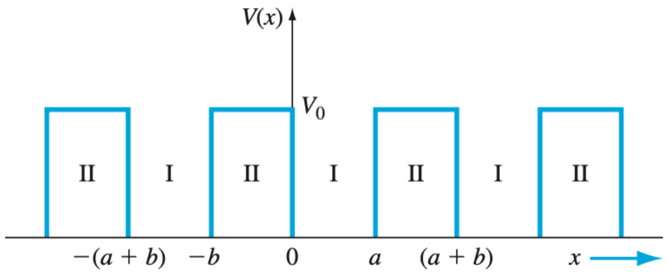
$$\frac{d^2\psi}{dx^2} = \left(\frac{d^2u}{dx^2} + 2ik\frac{du}{dx} - k^2 u(x) \right) e^{ikx}.$$

$$\alpha^2 = \frac{2mE}{\hbar^2}, \quad \beta^2 = \frac{2m(E - V_0)}{\hbar^2}.$$

$$\frac{d^2u}{dx^2} - 2ik\frac{du}{dx} - (k^2 - \frac{2m}{\hbar^2}(E - V(x)))u(x) = 0.$$



$$V(x) = \begin{cases} 0, & -b < x < 0, \\ V_0, & 0 < x < a, \end{cases}$$



Define $\alpha = \sqrt{\frac{2mE}{\hbar^2}}$. Then the solution is:

$$u_1(x) = Ae^{i(\alpha-k)x} + Be^{-i(\alpha+k)x}.$$

Region I: $0 < x < a$, **where** $V(x) = 0$

$$\frac{d^2u_1(x)}{dx^2} - 2ik\frac{du_1(x)}{dx} - (\alpha^2 - k^2)u_1(x) = 0.$$

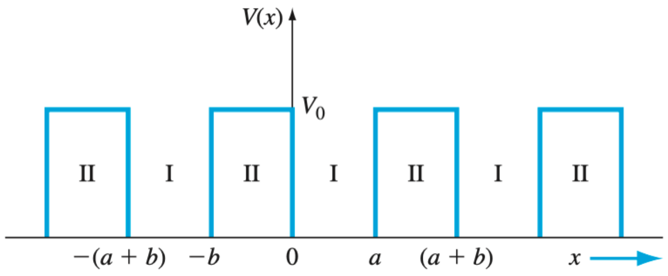
Region II: $-b < x < 0$, **where** $V(x) = V_0$

$$\frac{d^2u_2(x)}{dx^2} - 2ik\frac{du_2(x)}{dx} - (\beta^2 - k^2)u_2(x) = 0.$$

For $E < V_0$, define $\beta = \sqrt{\frac{2m(V_0-E)}{\hbar^2}}$. The solution becomes:

$$u_2(x) = Ce^{-i(\beta-k)x} + De^{i(\beta+k)x}.$$

$$V(x)=\begin{cases}0, & -b<x<0,\\ V_0, & 0<x<a,\end{cases}$$



Applying boundary conditions: continuity of wavefunction and derivative

At $x=0$:

$$u_1(0)=u_2(0)\Rightarrow A+B=C+D,$$

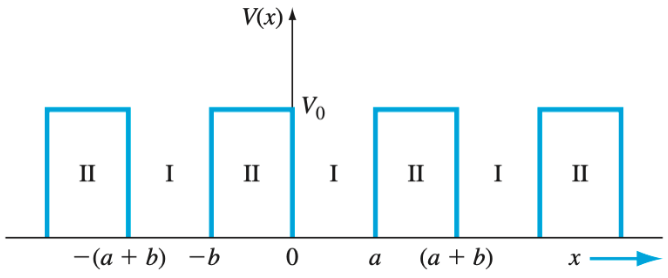
$$\left.\frac{du_1}{dx}\right|_{x=0}=\left.\frac{du_2}{dx}\right|_{x=0}\Rightarrow(\alpha-k)A-(\alpha+k)B=-(\beta-k)C+(\beta+k)D.$$

At $x=a$, recall that $u_1(a)=u_2(-b)$ due to periodicity:

$$u_1(a)=u_2(-b)\Rightarrow Ae^{i(\alpha-k)a}+Be^{-i(\alpha+k)a}=Ce^{i(\beta-k)b}+De^{-i(\beta+k)b},$$

$$\left.\frac{du_1}{dx}\right|_{x=a}=\left.\frac{du_2}{dx}\right|_{x=-b}\Rightarrow(\alpha-k)Ae^{i(\alpha-k)a}-(\alpha+k)Be^{-i(\alpha+k)a}$$
$$=(\beta-k)Ce^{i(\beta-k)b}-(\beta+k)De^{-i(\beta+k)b}.$$

$$V(x) = \begin{cases} 0, & -b < x < 0, \\ V_0, & 0 < x < a, \end{cases}$$



Beta disappear

Solving the system of boundary equations leads to a transcendental equation:

$$\cos(ka) = f(a),$$

In the limit where $V_0 \to \infty$, the expression becomes more compact (see delta barrier approximation):

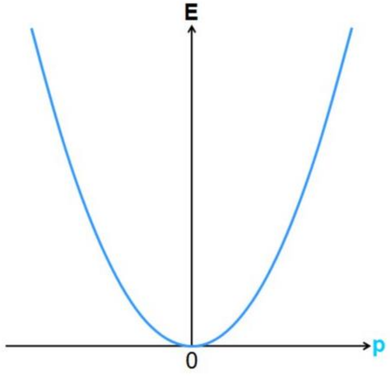
$$\cos(ka) = \cos(\alpha a) + \frac{mP}{\hbar^2 \alpha} \sin(\alpha a), \qquad \textit{Relation } E(k)$$

with $P = V_0(a - b)$.

We may note that Equation is ***not a solution of Schrodinger's*** *wave equation but gives* ***the conditions for which Schrodinger's wave equation will have a solution***. If we assume that the crystal is infinitely large, then *k* in Equation can assume a continuum of values and must be real.

$$\alpha = \sqrt{\frac{2mE}{\hbar^2}} = \frac{p}{\hbar} = k$$

$$E = \frac{p^2}{2m} = \frac{k^2\hbar^2}{2m}$$



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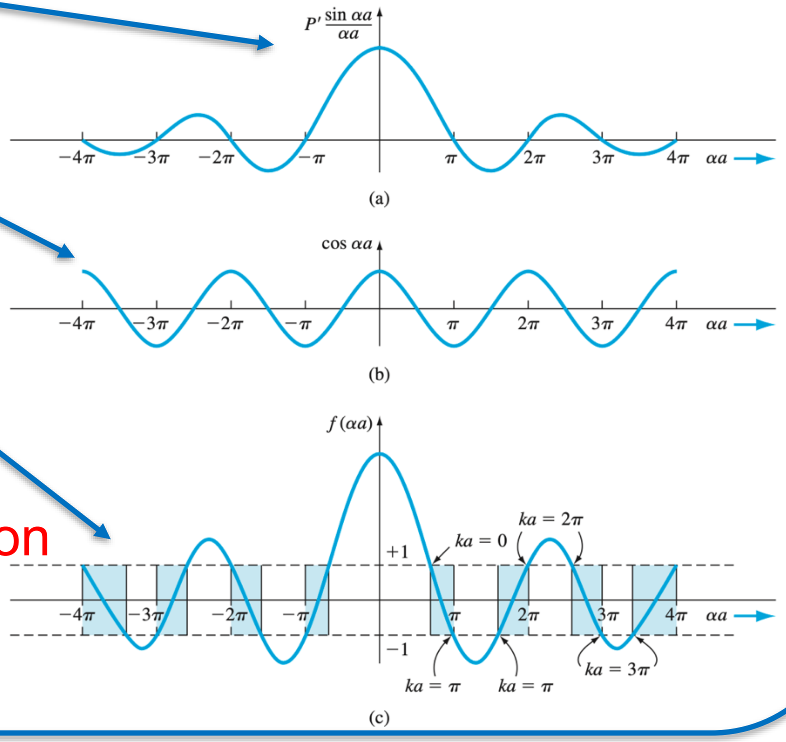
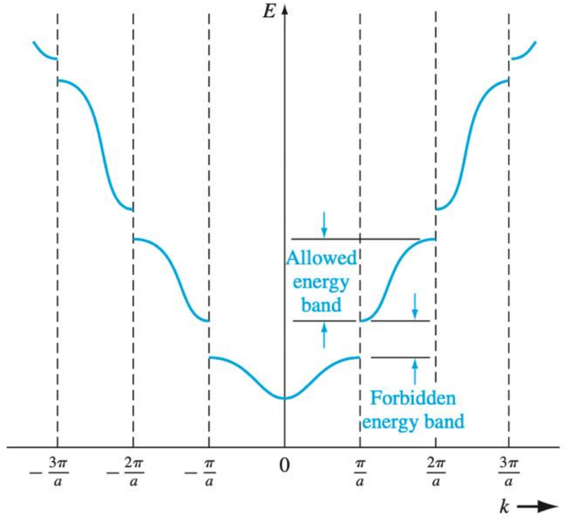
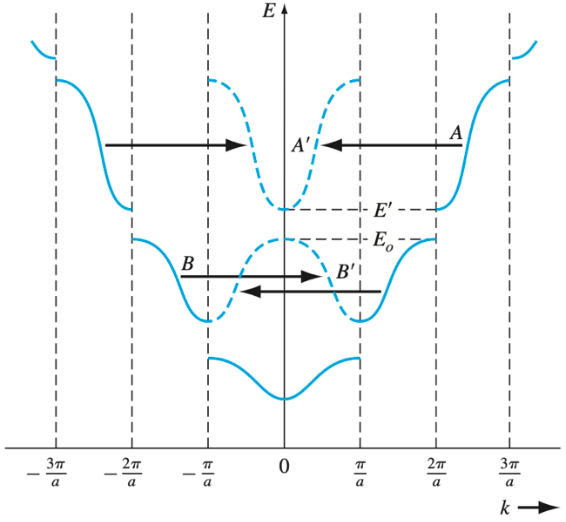
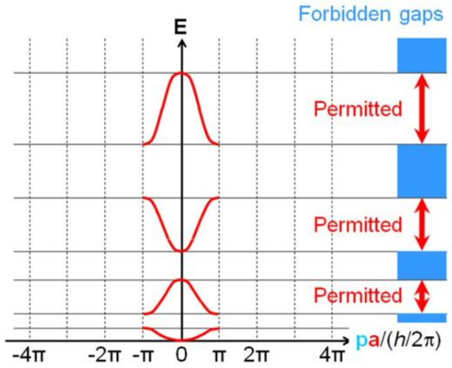


$$|\cos(ka)| \leq 1$$

Condition on the existence of solution







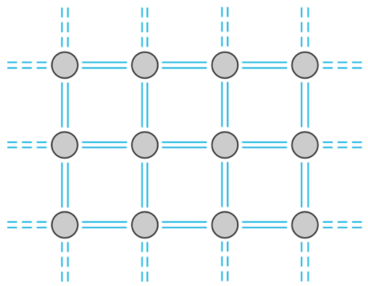
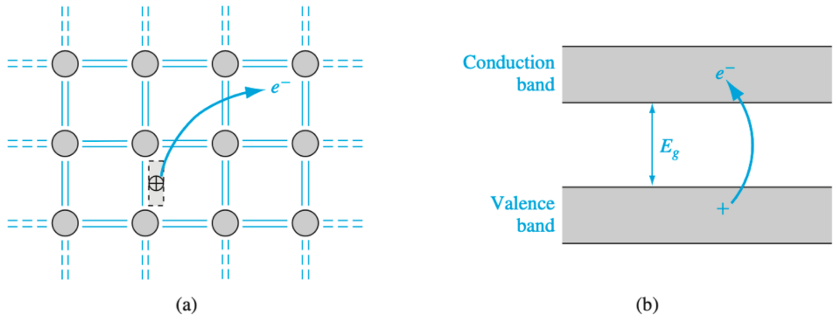
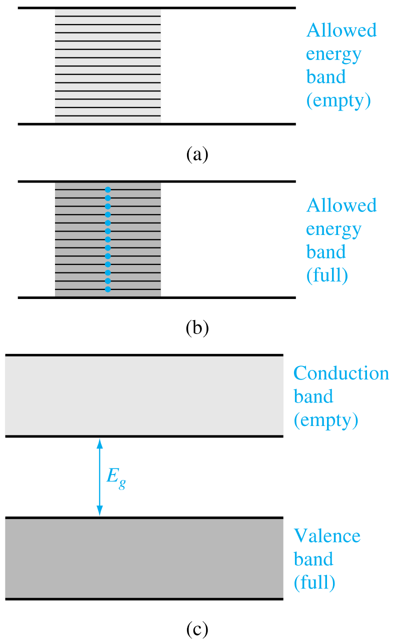
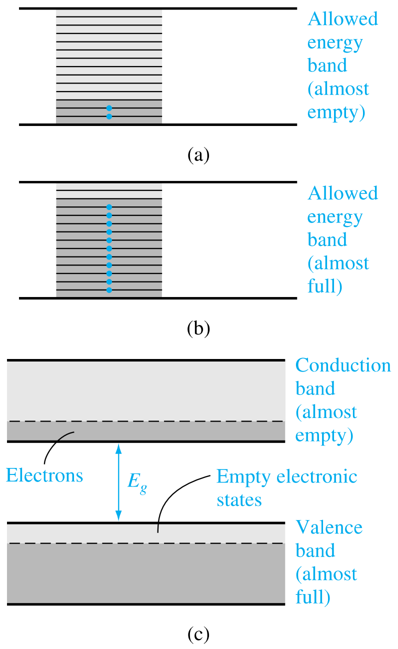


Figure 3.12 | Two-dimensional representation of the covalent bonding in a semiconductor at $T = 0$ K.

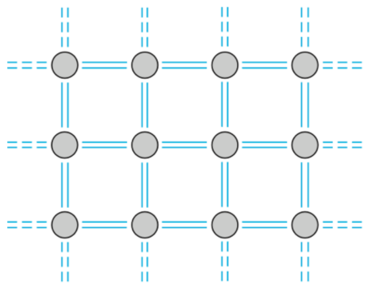
$$\psi_k(x) = e^{ikx}u_k(x)$$







- **Pauli exclusion principle**: In a completely filled band, for every electron with momentum +k, there's one with -k. The net electron motion (current) is zero.
- Only electrons in **partially filled bands** (like the conduction band, or near the top of a partially empty valence band) can change their state under an external field, creating a net current.
- Therefore, conduction occurs when:
- There are **free electrons** in the conduction band (e.g., in metals or n-type semiconductors), or
- There are **holes** (missing electrons) in the valence band (as in p-type semiconductors).



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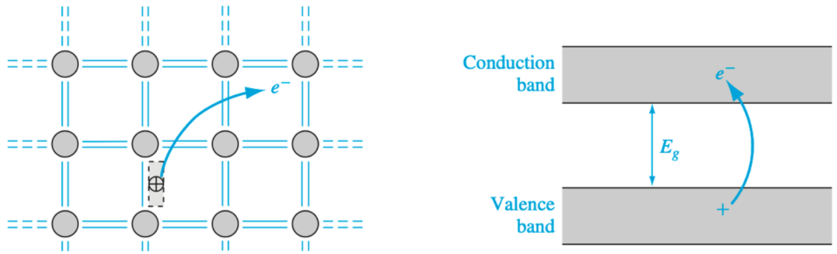
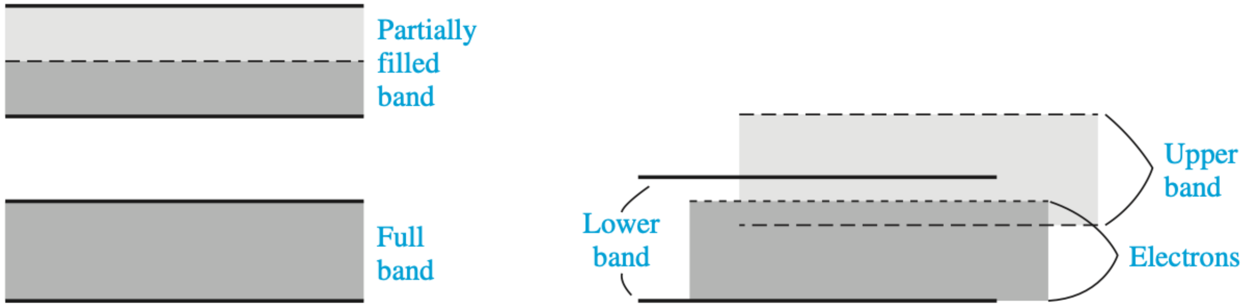
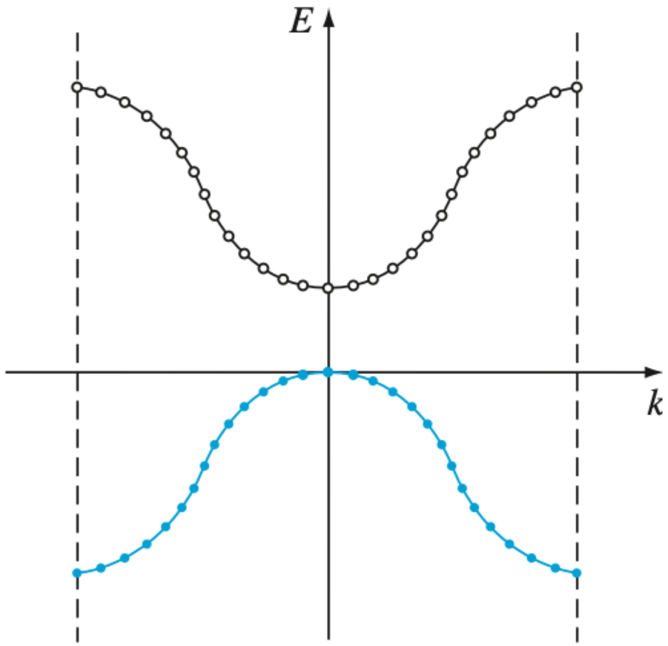


(a) (b)



(a) (b)

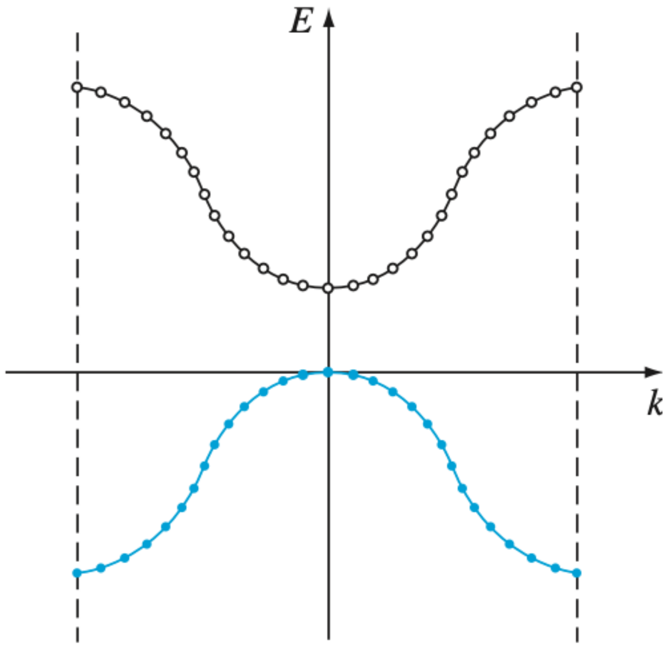
Two possible energy bands of a metal showing
(a) a partially filled band and
(b) overlapping allowed energy bands.



$$E = \frac{p^2}{2m}, \quad \text{with } p = \hbar k \Rightarrow E(k) = \frac{\hbar^2 k^2}{2m}$$

$$\frac{\mathrm{d}E}{\mathrm{d}k} = \frac{\hbar^2 k}{m} = \hbar \cdot \frac{\hbar k}{m} = \hbar v \quad \Rightarrow \quad \boxed{\frac{\mathrm{d}E}{\mathrm{d}k} = \hbar v} \quad \text{or} \quad \boxed{v = \frac{1}{\hbar}\frac{\mathrm{d}E}{\mathrm{d}k}}$$

$$\frac{\mathrm{d}^2 E}{\mathrm{d}k^2} = \frac{\hbar^2}{m}$$



Electrons near the conduction band minimum:

$$E(k) = E_c + C_1 k^2 \Rightarrow \frac{\mathrm{d}^2 E}{\mathrm{d}k^2} = 2C_1 \Rightarrow m_n^* = \frac{\hbar^2}{2C_1}$$

Holes near the valence band maximum:

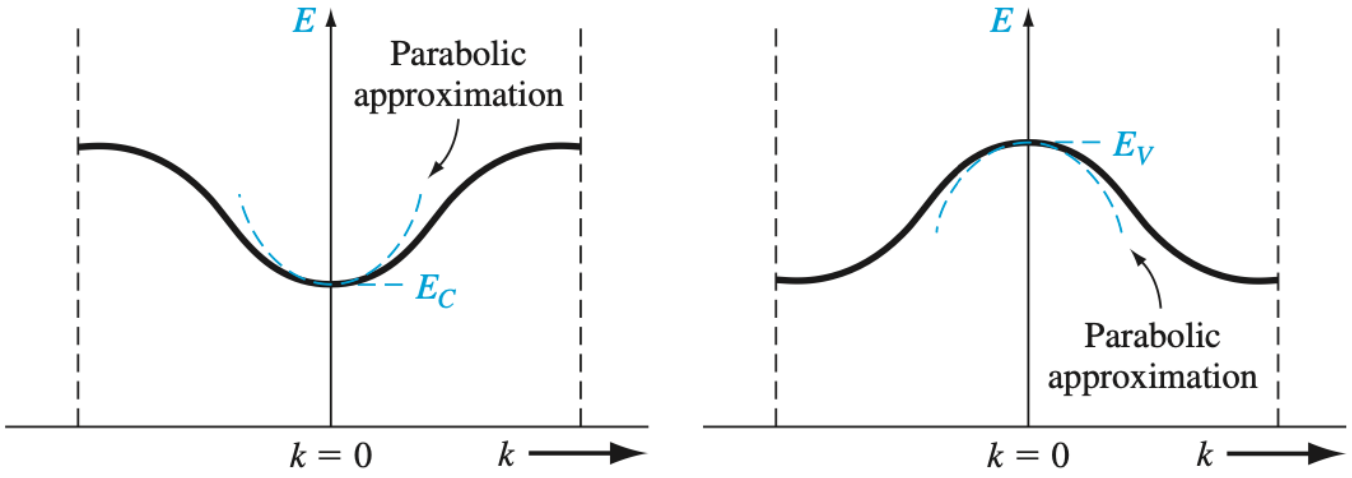
$$E(k) = E_v - C_2 k^2 \Rightarrow \frac{\mathrm{d}^2 E}{\mathrm{d}k^2} = -2C_2 \Rightarrow m_p^* = \frac{\hbar^2}{-2C_2} < 0$$

To describe conduction by holes, we define a positively charged quasi-particle with a positive mass:

$$m_p^* = -m_e^*$$

Unified definition:

$$\boxed{m^* = \hbar^2 \left(\frac{\mathrm{d}^2 E}{\mathrm{d}k^2}\right)^{-1}}$$



(a) (b)

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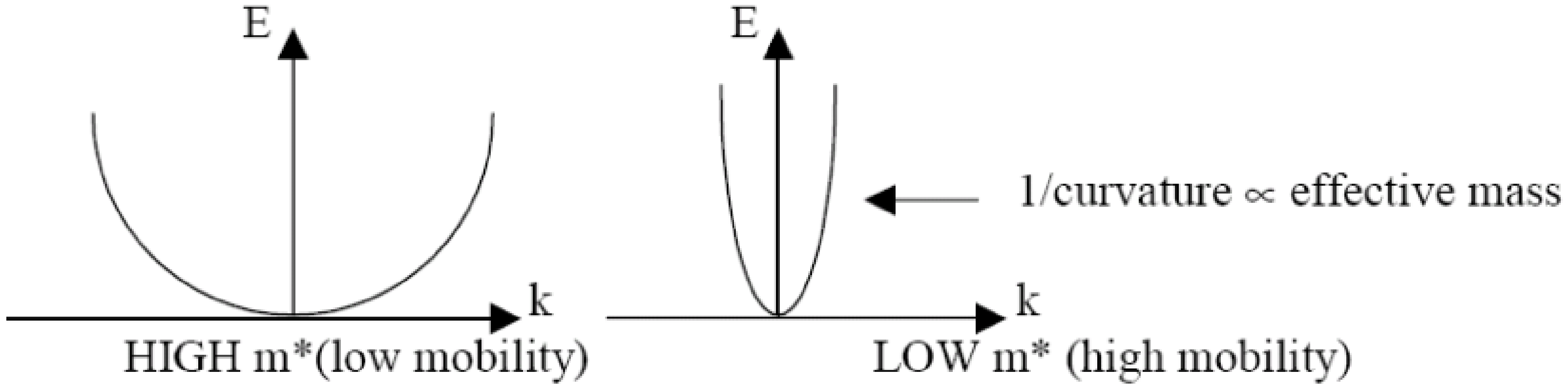


In 3 dimensions

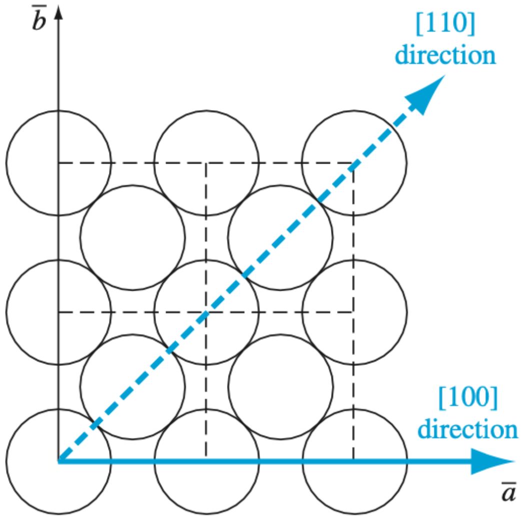


Figure 3.24 | The (100) plane of a face-centered cubic crystal showing the [100] and [110] directions.

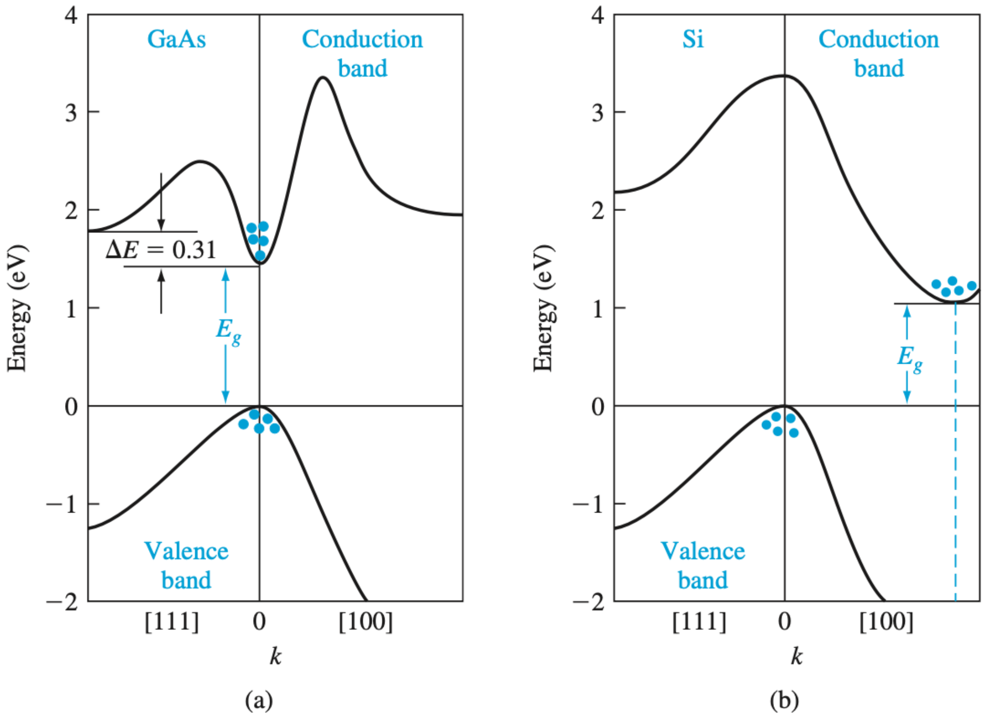
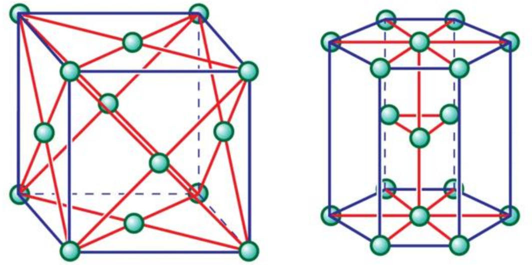
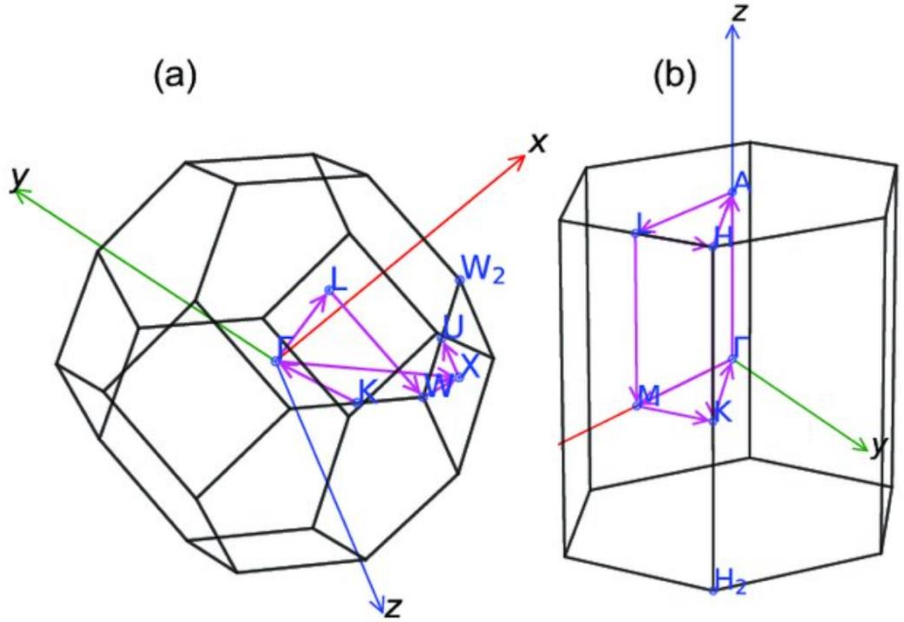
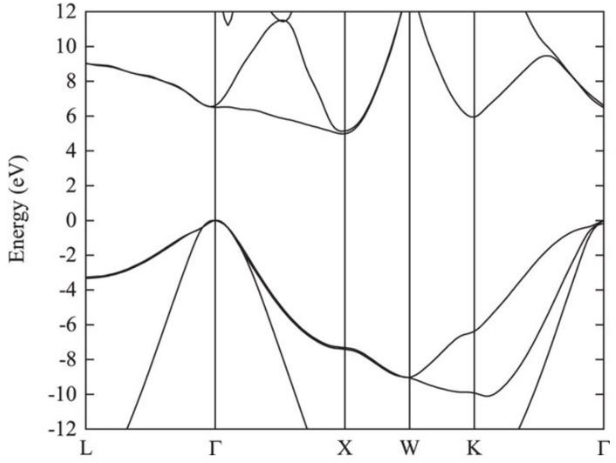


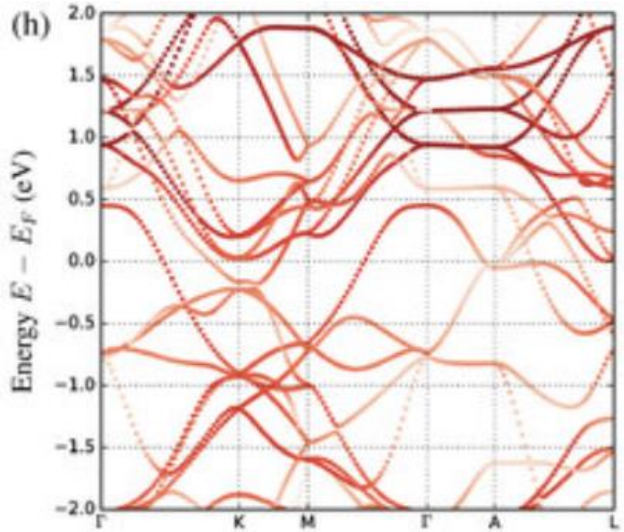
Figure 3.25 | Energy-band structures of (a) GaAs and (b) Si. *(From Sze [12].)*

In 3 dimensions



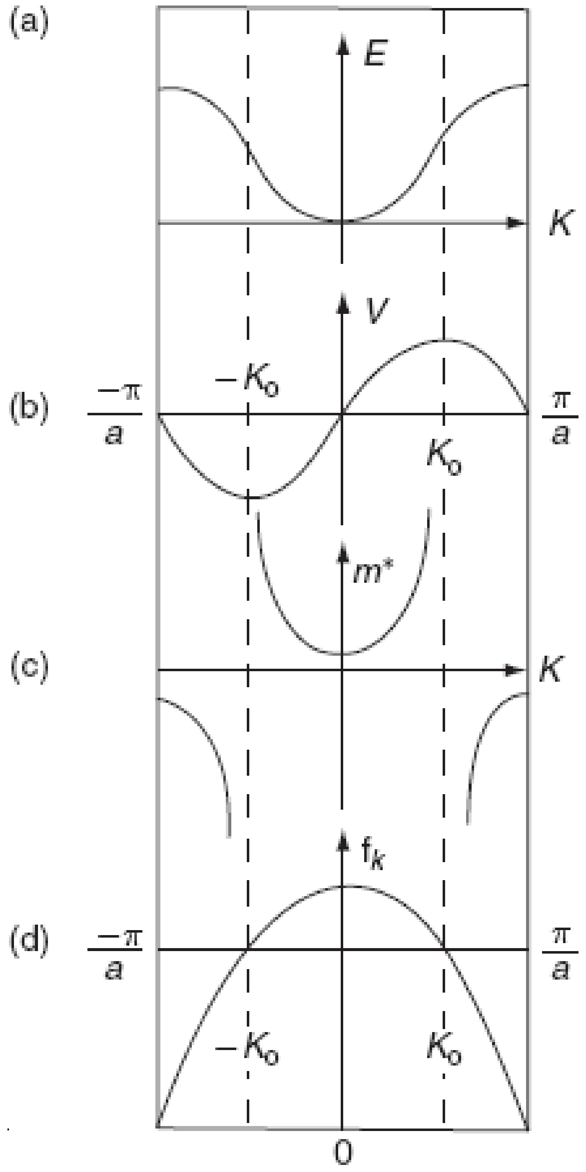






First Brillouin zone, special high symmetry points, and recommended k-paths for (a) face-centered cubic and hexagonal close packed lattices respectively.

Velocity, Acceleration, Effective Mass, and Force



$$E = \frac{\hbar^2 k^2}{2m^*}$$

$$v = \frac{\hbar k}{m^*} = \frac{1}{\hbar}\frac{dE}{dk}$$

$$m^* = \hbar^2 / \frac{d^2E}{dk^2}$$

$$F_{total} = F_{ext} + F_{int} = ma$$

$$F_{ext} = m^* a \qquad F_{int} = (m - m^*)a$$

$$F_{ext} dt = m^* dv = \hbar dk$$

$$F_{ext} = \hbar \frac{dk}{dt}$$

Free Electrons Vs Crystal Electrons

Free electrons

Wave function: $\psi_k(r) = \frac{1}{\sqrt{V}} e^{ikr}$

Kinetic energy: $E = \frac{\hbar^2 k^2}{2m}$

Velocity or group velocity:

$$\bar{v} = \int \psi^* \left(-\frac{i\hbar}{m} \nabla \right) \psi dr = \frac{\hbar k}{m}$$

Dynamics (F – force):

$$\frac{dv}{dt} = \frac{1}{m} F$$

Force equation:

$$F = \frac{dp}{dt} = \hbar \frac{dk}{dt}$$

Electrons in solid

Wave function: $\psi_k(r) = e^{ikr} u_k(r)$

Dispersion near band extremum (isotropic and parabolic): $E = \frac{\hbar^2 (k - k_0)^2}{2m^*}$

Group velocity: $v = \frac{1}{\hbar} \nabla_k E(k)$

Velocity at band extremum: $v = \frac{\hbar (k - k_0)}{m^*}$

Dynamics in a band:

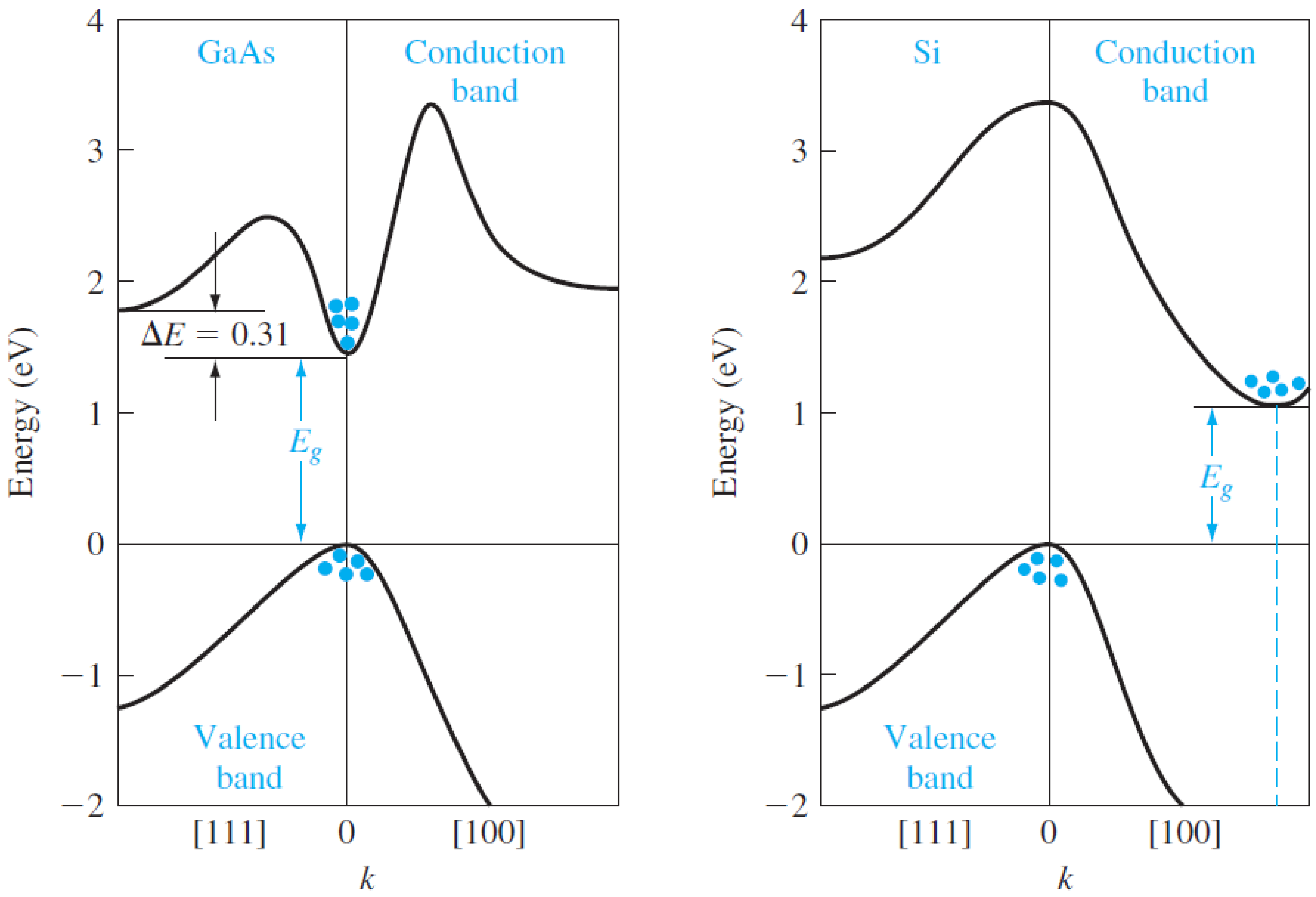
$$\frac{dv}{dt} = \frac{1}{\hbar} \nabla_k \frac{dE}{dt} = \frac{1}{\hbar} \nabla_k (Fv) = \frac{1}{\hbar^2} (\nabla_k \nabla_k E) F$$

$$\frac{dv}{dt} = \frac{1}{m^*} F; \quad \frac{1}{m^*} = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k^2}$$ (if m* isotropic and parabolic)

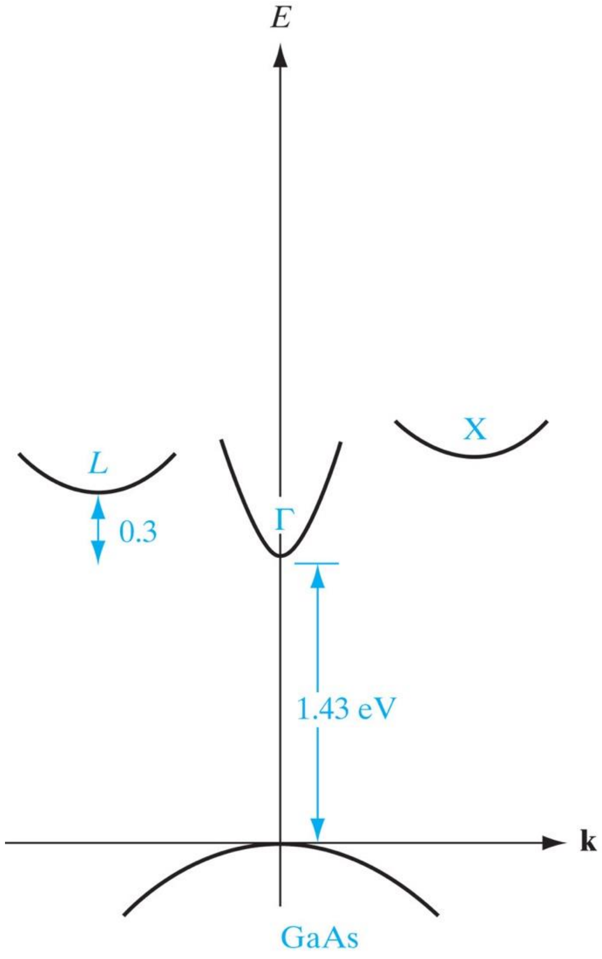
Force equation:

$$\frac{dE(k)}{dt} = \nabla_k E \frac{dk}{dt} = Fv \qquad F = \hbar \frac{dk}{dt}$$

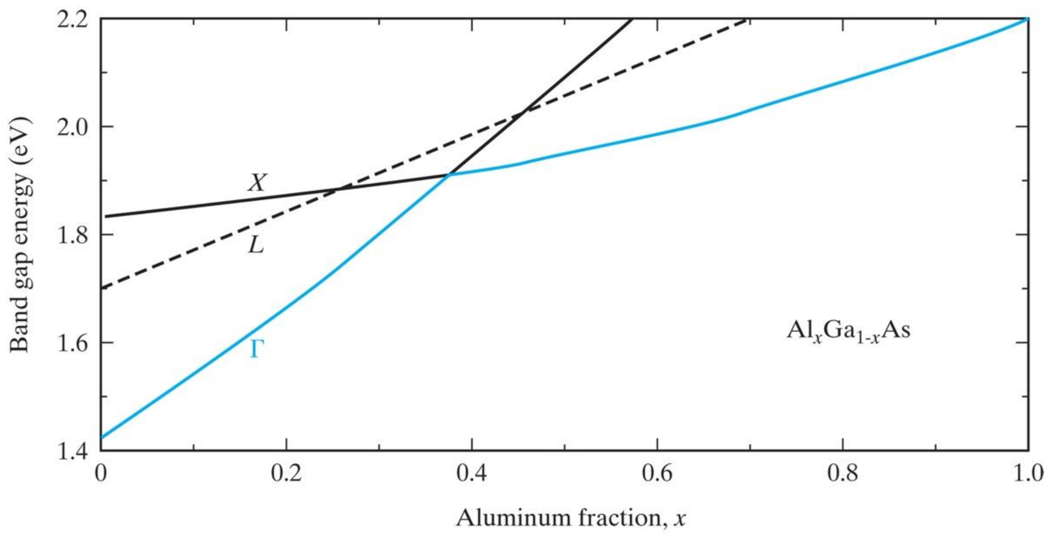
Indirect Bandgap and Direct Bandgap



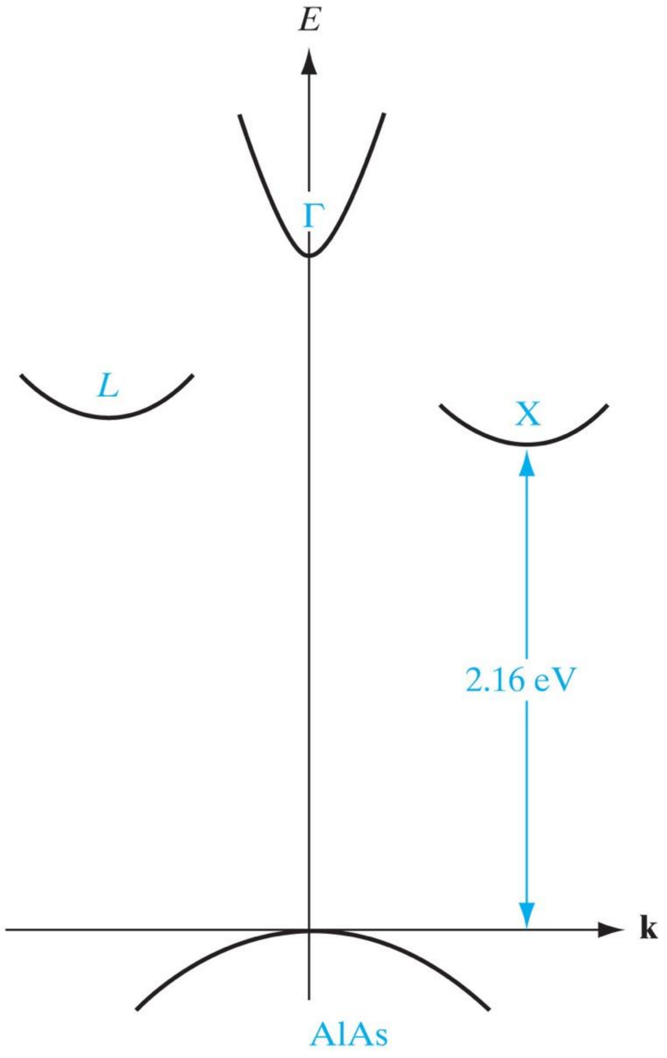
Band Engineering



(a)

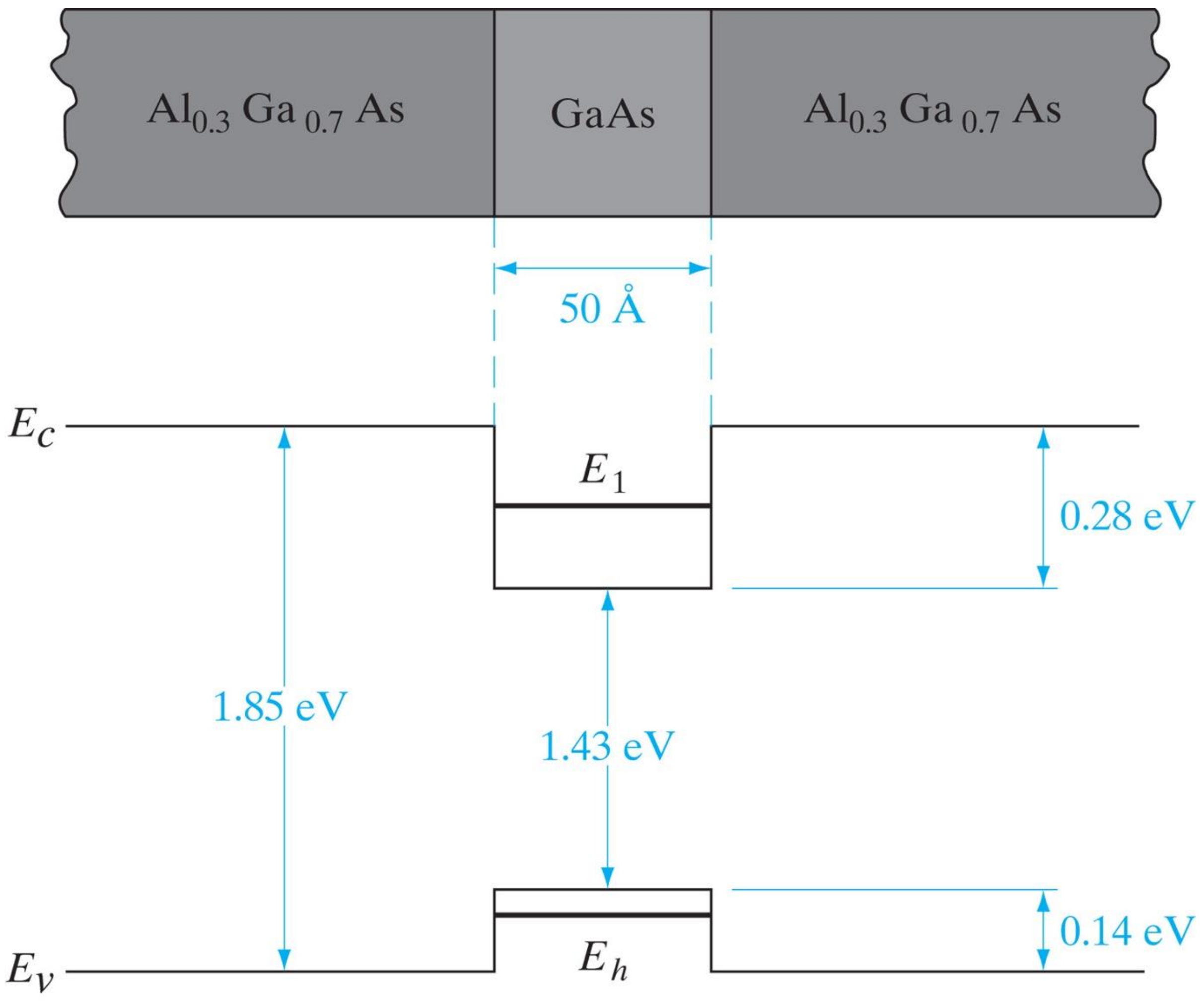


(c)



(b)

Quantum Well



Band Diagram & E-k Diagram

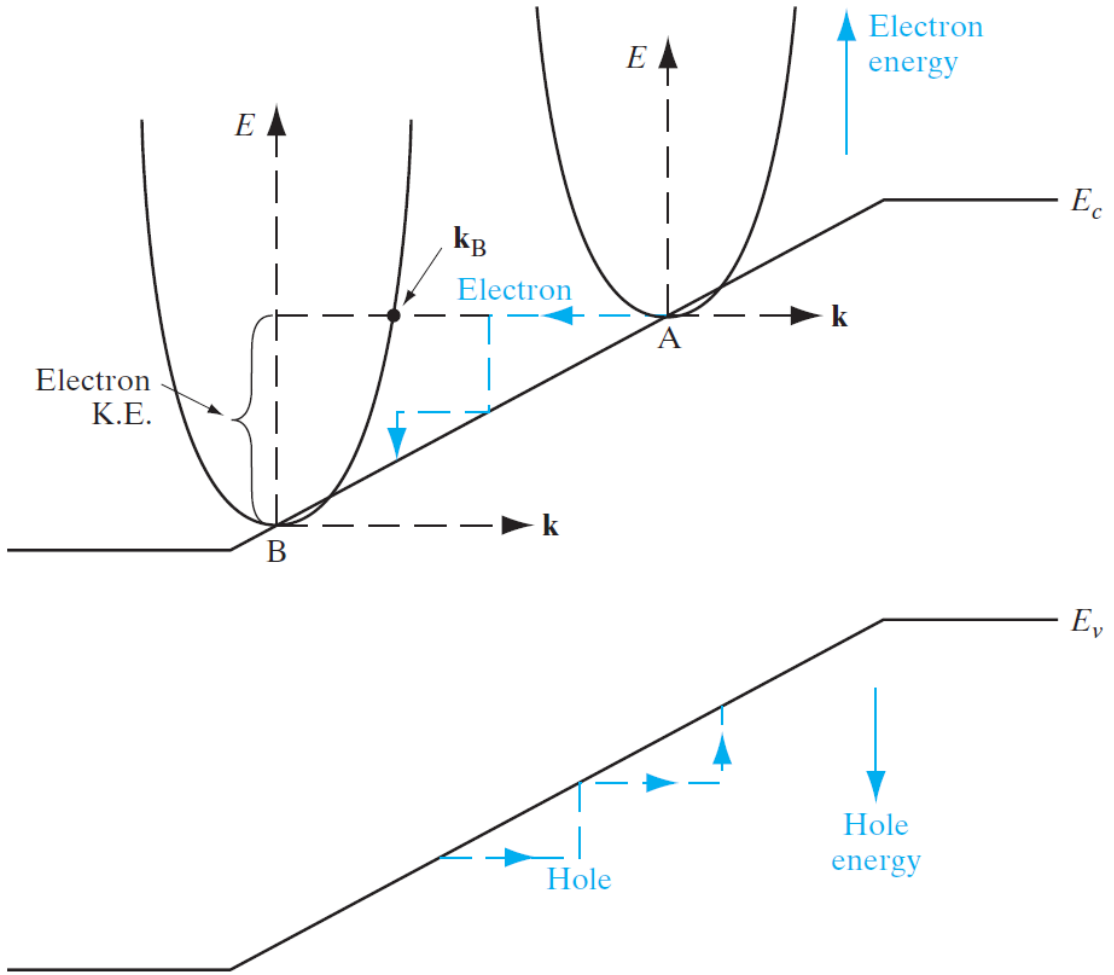
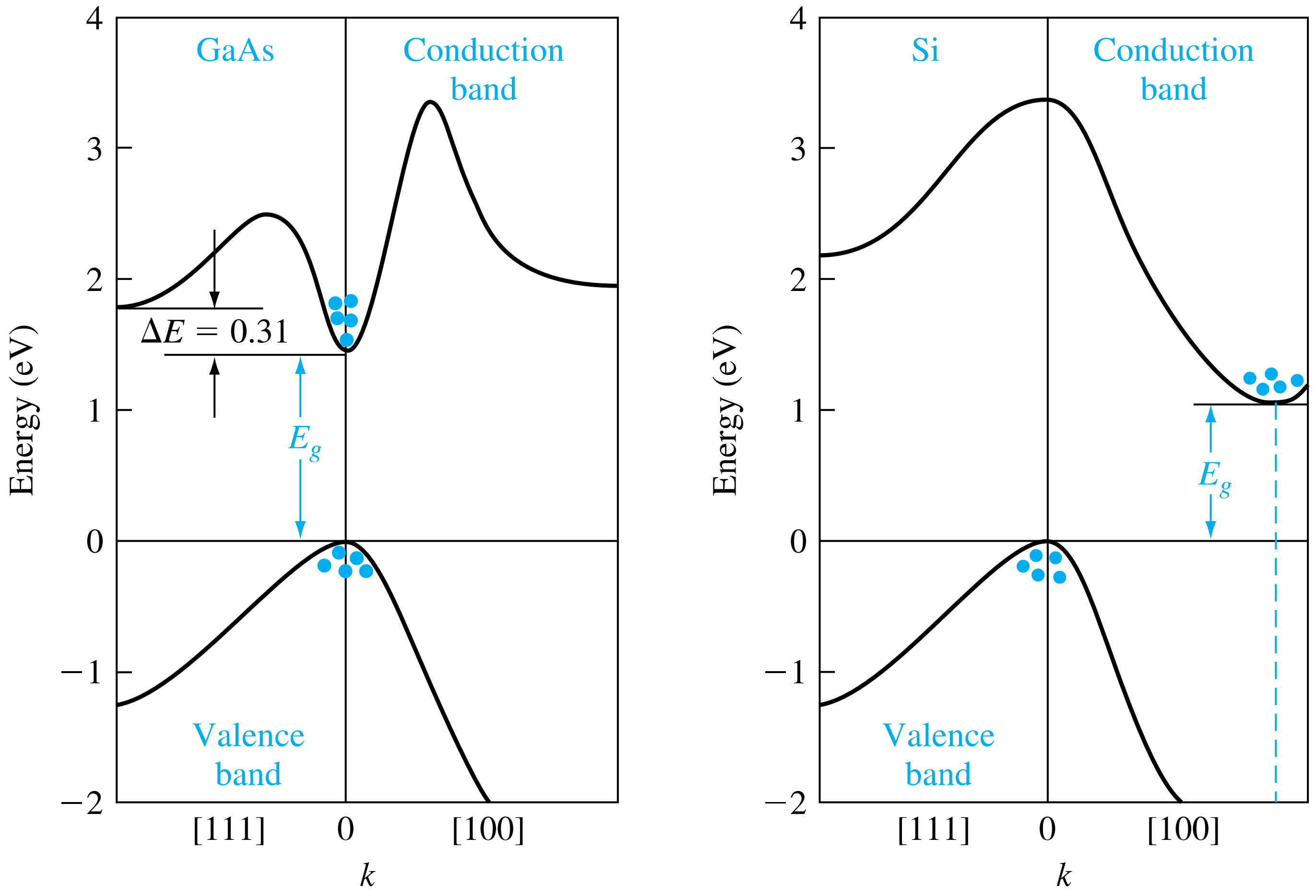


Figure 3–9
Superimposition of the (E, **k**) band structure on the E-versus-position simplified band diagram for a semiconductor in an electric field. Electron energies increase going up, while hole energies increase going down. Similarly, electron and hole wavevectors point in opposite directions and these charge carriers move opposite to each other, as shown.

Direct and indirect band gap



Quasi Particles

Interactions between electrons and the lattice (phonons, defects) are complex. Rather than modeling these in detail, we use the concept of a **quasi-particle** an entity that behaves like a particle but with modified properties:

- Modified energy $E(k)$
- Modified mass m^*
- Finite lifetime (due to scattering)

Electrons and holes are the primary quasi-particles in band theory. They allow us to treat solid-state systems with powerful classical analogies.